

Name: _____

Use the Base Angles Theorem and its converse.

1. The vertex angle of an isosceles triangle measures 52° . Find each base angle.
2. A base angle of an isosceles triangle measures 37° . Find the vertex angle.
3. The base angles of a triangle measure $(2x + 13)^\circ$ and $(4x - 9)^\circ$. Find x and each base angle.
4. Each angle of an equilateral triangle is written as $(5y - 5)^\circ$. Find y .

Use the side lengths.

5. The legs of an isosceles triangle measure $3x + 4$ and $5x - 12$. Find x and the length of each leg.
6. That triangle has a base of 18. Find its perimeter.

Answer each question.

7. In $\triangle DEF$, $m\angle D = 68^\circ$ and $m\angle E = 68^\circ$. What can you conclude about the sides? Name the theorem.
8. A triangle has angles measuring 60° , 60° , and 60° . Classify it two ways and justify.
9. In the diagram, $\overline{AB} \cong \overline{AC}$, $\overline{BD} \cong \overline{BC}$, and $m\angle A = 38^\circ$. Find $m\angle DBC$.
10. Explain why the vertex angle of an isosceles triangle can never measure 180° or more.

